



Metropolitan University, Sylhet

Department of Computer Science & Engineering

Lab Report – 03:

Coin Change - Maximum Coin Combinations (Dynamic Programming)

Course Title: Algorithm Design and Analysis Lab

Course Code: CSE 232

Submitted To:

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Batch: CSE 62nd (A)

Date of Submission:

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problem:

Coin Change - Maximum Coin Combinations (Dynamic Programming)

CODE :

```
// Coin Change - Maximum Coin Combinations (Dynamic Programming)
// Name: Mahdi Hasan Shuvo | ID: 251-115-030 | Batch: CSE 62nd (A)

#include<iostream>
#include<vector>

using namespace std;
int CoinChange(int n, vector <int> &coins, int amount)
{
    vector <vector <int>> Table (n+1, vector <int> (amount+1, 0));

    for (int i = 0; i <= n; i++)
    {
        Table[i][0] = 1;
    }

    for (int i = 1; i <= n; i++)
    {
        for (int a = 1; a <= amount; a++)
        {
            if (coins[i-1] <= a)
            {
                Table[i][a] = Table[i-1][a] + Table[i][a-coins[i-1]];
            }

            else
                Table[i][a] = Table[i-1][a];
        }
    }

    return Table[n][amount];
}

int main()
{
    int n;
```

```

    cout << "Enter number of coins: ";
    cin >> n;

    vector <int> coins(n);

    for (int i = 0; i < n; i++)
    {
        cout << "Coin-" << i+1 << ": ";
        cin >> coins[i];
    }

    int amount;
    cout << "Enter amount: ";
    cin >> amount;

    int combination = CoinChange(n, coins, amount);
    cout << "Total number of combination is " << combination << endl;

    return 0;
}

```

Output Output Output

OUTPUT:

The screenshot shows a C++ IDE with a file named "Coin Change (Maximum coin combination).cpp". The code is as follows:

```

1 // Coin Change - Maximum Coin Combinations (Dynamic Programming)

```

The IDE has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the following output:

```

Enter number of coins: 5
• Coin-1: 2
  Coin-2: 3
  Coin-3: 4
  Coin-4: 5
  Coin-5: 6
Enter amount: 78
Total number of combination is 3472
○ PS G:\CSE-62\AD&A>

```

